ENGINE COOLANT THERMOSTAT

DESCRIPTION

CAUTION: Do not operate an engine without a thermostat, except for servicing or testing.

The thermostat (Fig. 8) on all gas powered engines is located beneath the thermostat housing at the front of the intake manifold.

The thermostat is a wax pellet driven, reverse poppet choke type.

Coolant leakage into the pellet container will cause the thermostat to fail in the open position. Thermostats very rarely stick. Do not attempt to free a thermostat with a prying device.

The same thermostat is used for winter and summer seasons. An engine should not be operated without a thermostat, except for servicing or testing. Operating without a thermostat causes longer engine warm-up time, unreliable warm-up performance, increased exhaust emissions and crankcase condensation that can result in sludge formation.

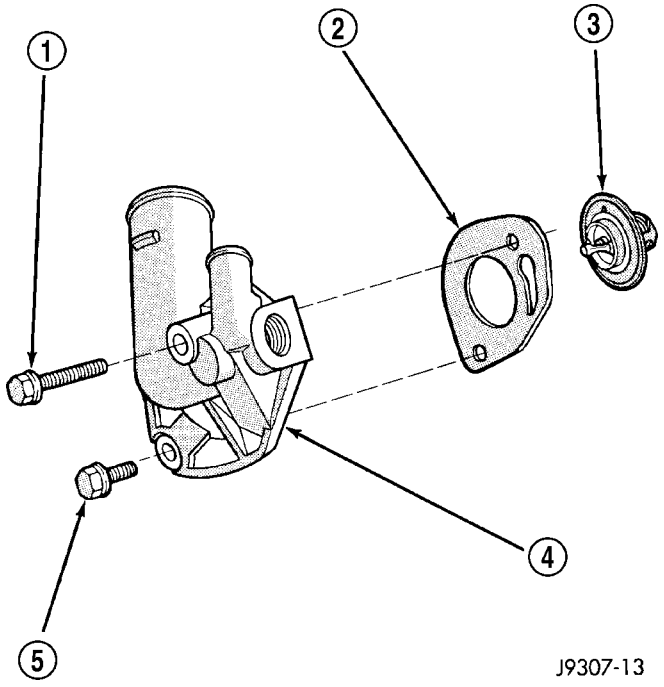

Fig. 8 Thermostat—Typical

1 - LONG BOLT
2 - GASKET
3 - THERMOSTAT
4 - THERMOSTAT HOUSING
5 - SHORT BOLT

OPERATION

The wax pellet is located in a sealed container at the spring end of the thermostat. When heated, the pellet expands, overcoming closing spring tension and water pump pressure to force the valve to open.

REMOVAL

2.4L

(1) Drain cooling system below thermostat housing level.
(2) Disconnect engine coolant temperature sensor.
(3) Disconnect heater supply hose.
(4) Remove housing attaching bolts (Fig. 9).
(5) Remove housing, gasket and thermostat (Fig. 9).

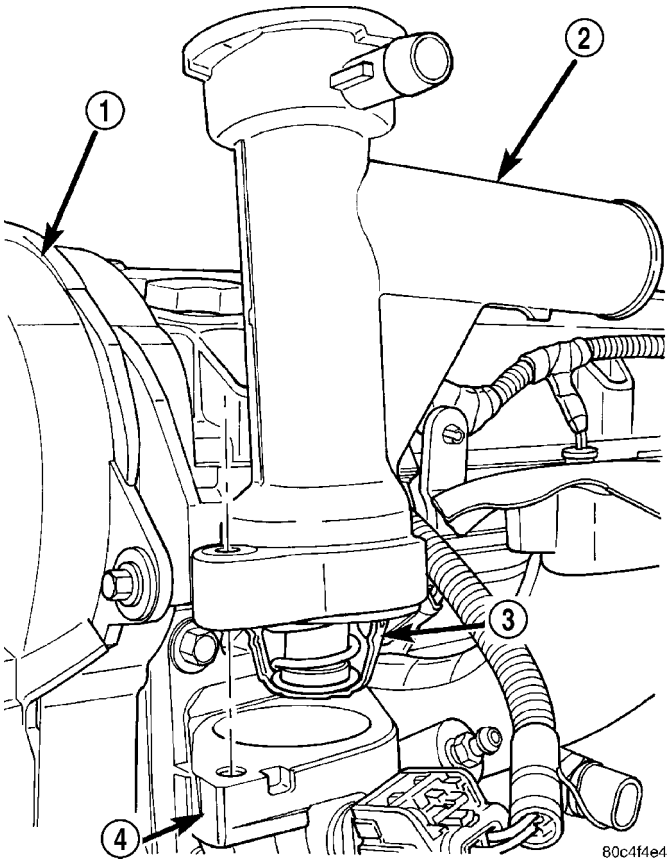

Fig. 9 Thermostat and Coolant Outlet Connector

1 - TIMING BELT COVER
2 - OUTLET CONNECTOR
3 - THERMOSTAT
4 - HOUSING

4.0L

WARNING: Do not loosen the radiator draincock with the system hot and pressurized. serious burns from the coolant can occur.

DO NOT WASTE reusable coolant. If the solution is clean, drain the coolant into a clean container for reuse.

(1) Drain the coolant from the radiator until the level is below the thermostat housing (Refer to 7 - COOLING - STANDARD PROCEDURE).